

高等微积分

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第六章: 求导的逆运算-不定积分





普通高等教育“十五”国家级规划教材

数学分析教程

(上册)

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§ 6.1. 不定积分的概念与积分法

定义 1. 将定义在区间上的函数 f 的原函数的一般表达式称为 f 的不定积分, 记作 $\int f(x) dx$.

评注: (1) $\int f(x) dx$ 是以 x 为自变量的函数.

(2) 若 F, G 均为 f 的原函数, 则 $F' = G'$, 于是存在常数 C 使得 $G - F = C$, 故

$$\int f(x) dx = F(x) + C \text{ (其中 } C \text{ 为常数)}.$$

(3) 若 $f \in \mathcal{C}[a, b]$, 则 $\int f(x) dx = \int_a^x f(t) dt + C$.

不定积分与导数、微分的关系

• 若 $\int f(x) dx = F(x) + C$, 则 $F'(x) = f(x)$,

$$\left(\int f(x) dx \right)' = F'(x) = f(x),$$

$$dF(x) = f(x) dx, \quad d\left(\int f(x) dx \right) = f(x) dx,$$

$$\int f(x) dx = \int F'(x) dx = \int dF(x) = F(x) + C.$$

- (线性性) $\forall \alpha, \beta \in \mathbb{R}$, 我们有

$$\int (\alpha f(x) + \beta g(x)) \, dx = \alpha \int f(x) \, dx + \beta \int g(x) \, dx.$$

基本的不定积分公式

- $\int dx = x + C.$
- $\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C \ (\alpha \neq -1),$
- $\int \frac{1}{x} dx = \ln |x| + C.$
- $\int a^x dx = \frac{a^x}{\ln a} + C \ (a > 0, \ a \neq 1),$
- $\int e^x dx = e^x + C.$
- $\int \sin x dx = -\cos x + C, \int \cos x dx = \sin x + C.$

- $\int \operatorname{sh} x \, dx = \operatorname{ch} x + C, \int \operatorname{ch} x \, dx = \operatorname{sh} x + C.$
- $\int \sec^2 x \, dx = \tan x + C.$
- $\int \csc^2 x \, dx = -\cot x + C.$
- $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C.$
- $\int \frac{dx}{1+x^2} = \arctan x + C.$
- $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln |x + \sqrt{x^2+a^2}| + C.$
- $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln |x + \sqrt{x^2-a^2}| + C.$

计算不定积分的基本方法

例 1. 计算 $\int |x - 1| dx$.

解: 当 $x \geq 1$ 时, 我们有

$$\int |x - 1| dx = \int (x - 1) dx = \frac{1}{2}x^2 - x + C_1.$$

当 $x < 1$ 时, 我们则有

$$\int |x - 1| dx = \int (1 - x) dx = x - \frac{1}{2}x^2 + C_2.$$

由于原函数在点 $x = 1$ 处连续, 则 $C_1 = 1 + C_2$,

$$\text{故 } \int |x - 1| dx = \begin{cases} \frac{1}{2}x^2 - x + 1 + C, & \text{若 } x \geq 1, \\ x - \frac{1}{2}x^2 + C, & \text{若 } x < 1. \end{cases}$$

例 2. 计算 $\int \frac{2x^2}{1+x^2} dx$.

解:
$$\begin{aligned}\int \frac{2x^2}{1+x^2} dx &= 2 \int \left(1 - \frac{1}{1+x^2}\right) dx \\ &= 2 \int dx - 2 \int \frac{1}{1+x^2} dx \\ &= 2x - 2 \arctan x + C.\end{aligned}$$

例 3. 计算 $\int \frac{dx}{(\cos^2 x)(\sin^2 x)}.$

解:
$$\begin{aligned}\int \frac{dx}{(\cos^2 x)(\sin^2 x)} &= \int \frac{\cos^2 x + \sin^2 x}{(\cos^2 x)(\sin^2 x)} dx \\ &= \int \left(\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} \right) dx \\ &= \tan x - \cot x + C.\end{aligned}$$

例 4. 计算 $\int \left(3x^2 - \frac{1}{\sqrt[3]{x}} + \frac{1}{x} + \frac{1}{1+x^2}\right) dx$.

解:
$$\begin{aligned} & \int \left(3x^2 - \frac{1}{\sqrt[3]{x}} + \frac{1}{x} + \frac{1}{1+x^2}\right) dx \\ &= \int d\left(x^3 - \frac{3}{2}x^{\frac{2}{3}} + \ln|x| + \arctan x\right) \\ &= x^3 - \frac{3}{2}x^{\frac{2}{3}} + \ln|x| + \arctan x + C. \end{aligned}$$

例 5. 计算 $\int \frac{dx}{a^2-x^2}$ ($a \neq 0$).

解:
$$\begin{aligned}\int \frac{dx}{a^2-x^2} &= \int \frac{1}{2a} \left(\frac{1}{x+a} - \frac{1}{x-a} \right) dx \\ &= \frac{1}{2a} \left(\ln |x+a| - \ln |x-a| \right) + C \\ &= \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right| + C.\end{aligned}$$

例 6. 计算 $\int \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2 dx$.

解:
$$\begin{aligned} \int \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2 dx &= \int (1 + \sin x) dx \\ &= x - \cos x + C. \end{aligned}$$

例 7. 计算 $\int e^{|x|} dx$.

解: 当 $x \geq 0$, $\int e^{|x|} dx = \int e^x dx = e^x + C_1$.

当 $x < 0$ 时, $\int e^{|x|} dx = \int e^{-x} dx = -e^{-x} + C_2$.

由于原函数在点 $x = 0$ 连续, 从而 $C_2 = 2 + C_1$.

故 $\int e^{|x|} dx = \begin{cases} e^x + C, & \text{若 } x \geq 0, \\ 2 - e^{-x} + C, & \text{若 } x < 0. \end{cases}$

§ 6.2. 方法一: 第一换元积分法 (凑微分)

若 $F'(y) = f(y)$, 而 u 为可导函数, 则

$$(F \circ u)'(x) = F'(u(x))u'(x) = f(u(x))u'(x).$$

故 $\int f(u(x))u'(x)dx = F(u(x)) + C$. 通常也将左式写成 $\int f(u(x))du(x)$, 则我们有

$$\begin{aligned}\int f(u(x))u'(x)dx &= \int f(u(x))du(x) \\ &= F(u(x)) + C.\end{aligned}$$

例 8. 计算 $\int 2xe^{x^2} dx$.

解: $\int 2xe^{x^2} dx = \int e^{x^2} d(x^2) = \int d(e^{x^2}) = e^{x^2} + C$.

例 9. 设 $a \neq 0$. 计算 $\int \frac{dx}{a^2+x^2}$.

解: $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \int \frac{d(\frac{x}{a})}{1+(\frac{x}{a})^2} = \frac{1}{a} \int d(\arctan \frac{x}{a})$
 $= \frac{1}{a} \arctan \frac{x}{a} + C$.

例 10. 设 $a > 0$. 计算 $\int \frac{dx}{\sqrt{a^2 - x^2}}$.

解: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \int \frac{d(\frac{x}{a})}{\sqrt{1 - (\frac{x}{a})^2}} = \int d(\arcsin \frac{x}{a}) = \arcsin \frac{x}{a} + C.$

例 11. 计算 $\int \tan x \, dx$.

$$\begin{aligned}\text{解: } \int \tan x \, dx &= \int \frac{\sin x}{\cos x} dx = - \int \frac{d(\cos x)}{\cos x} \\ &= - \int d(\ln |\cos x|) = - \ln |\cos x| + C.\end{aligned}$$

例 12. 计算 $\int \cot x \, dx$.

$$\text{解: } \int \cot x \, dx = \int \frac{\cos x}{\sin x} dx = \int \frac{d(\sin x)}{\sin x} = \ln |\sin x| + C.$$

例 13. 计算 $\int \tan^2 x \, dx$.

解:
$$\begin{aligned}\int \tan^2 x \, dx &= \int (\sec^2 x - 1) \, dx \\ &= \int d(\tan x - x) = \tan x - x + C.\end{aligned}$$

例 14. 计算 $\int \frac{dx}{(1+4x^2)(\arctan 2x+1)^2}$.

解:
$$\begin{aligned} & \int \frac{dx}{(1+4x^2)(1+\arctan 2x)^2} \\ &= \frac{1}{2} \int \frac{d(2x)}{(1+(2x)^2)(1+\arctan 2x)^2} = \frac{1}{2} \int \frac{d(\arctan 2x)}{(1+\arctan 2x)^2} \\ &= \frac{1}{2} \int \frac{d(1+\arctan 2x)}{(1+\arctan 2x)^2} = -\frac{1}{2} \int d\left(\frac{1}{1+\arctan 2x}\right) \\ &= -\frac{1}{2(1+\arctan 2x)} + C. \end{aligned}$$

例 15. 计算 $\int \frac{dx}{\sin x}$.

$$\begin{aligned}\text{解: } \int \frac{dx}{\sin x} &= \int \frac{dx}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \int \frac{d(\frac{x}{2})}{\tan \frac{x}{2} \cos^2 \frac{x}{2}} \\ &= \int \frac{d(\tan \frac{x}{2})}{\tan \frac{x}{2}} = \ln \left| \tan \frac{x}{2} \right| + C.\end{aligned}$$

例 16. 计算 $\int \frac{x}{\sqrt{3+2x-x^2}} dx$.

解:
$$\begin{aligned} & \int \frac{x}{\sqrt{3+2x-x^2}} dx \\ &= \int \left(-\frac{1}{2} \frac{(3+2x-x^2)'}{\sqrt{3+2x-x^2}} + \frac{1}{\sqrt{3+2x-x^2}} \right) dx \\ &= -\frac{1}{2} \int \frac{d(3+2x-x^2)}{\sqrt{3+2x-x^2}} + \int \frac{1}{\sqrt{4-(x-1)^2}} dx \\ &= -\sqrt{3+2x-x^2} + \int \frac{1}{\sqrt{1-(\frac{x-1}{2})^2}} d\left(\frac{x-1}{2}\right) \\ &= -\sqrt{3+2x-x^2} + \arcsin \frac{x-1}{2} + C. \end{aligned}$$

例 17. 计算 $\int \frac{dx}{1+e^x}$.

解: $\int \frac{dx}{1+e^x} = -\int \frac{d(e^{-x})}{e^{-x}+1} = -\ln(1+e^{-x}) + C.$

例 18. 计算 $\int \sec x \, dx$.

$$\begin{aligned}\text{解: } \int \sec x \, dx &= \int \frac{dx}{\cos x} = \int \frac{d(x + \frac{\pi}{2})}{\sin(x + \frac{\pi}{2})} \\ &= \ln \left| \csc(x + \frac{\pi}{2}) - \cot(x + \frac{\pi}{2}) \right| + C \\ &= \ln |\sec x + \tan x| + C.\end{aligned}$$

§ 6.2 方法二: 第二换元积分法

$$\begin{aligned}\int f(x) \, dx &\stackrel{x=x(t)}{=} \int f(x(t)) x'(t) \, dt \\ &\stackrel{t=t(x)}{=} F(t(x)) + C.\end{aligned}$$

例 19. 计算 $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

解:
$$\begin{aligned} \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx &\stackrel{t=\sqrt{x}}{=} \int \frac{\sin t}{t} d(t^2) = \int \frac{\sin t}{t} \cdot (2t) dt \\ &= \int 2 \sin t dt = -2 \cos t + C \\ &= -2 \cos \sqrt{x} + C. \end{aligned}$$

例 20. 计算 $\int \frac{dx}{1+\sqrt[3]{1+x}}$.

解:
$$\begin{aligned} \int \frac{dx}{1+\sqrt[3]{1+x}} &\stackrel{u=\sqrt[3]{1+x}}{=} \int \frac{d(u^3-1)}{1+u} \\ &= \int \frac{3u^2 du}{1+u} = 3 \int \left(u - 1 + \frac{1}{1+u}\right) du \\ &= 3\left(\frac{1}{2}u^2 - u + \ln |1+u|\right) + C \\ &= 3\left(\frac{1}{2}(1+x)^{\frac{2}{3}} - \sqrt[3]{1+x} + \ln |1+\sqrt[3]{1+x}|\right) + C. \end{aligned}$$

例 21. 计算 $\int \frac{x-2}{1-\sqrt{x+1}} dx$.

解:
$$\begin{aligned} \int \frac{x-2}{1-\sqrt{x+1}} dx &\stackrel{u=\sqrt{x+1}}{=} \int \frac{u^2-3}{1-u} d(u^2-1) \\ &= \int \left(-2u^2 - 2u + 4 - \frac{4}{1-u} \right) du \\ &= -\frac{2}{3}u^3 - u^2 + 4u + 4 \ln |1-u| + C_1 \\ &= -\frac{2}{3}(x-5)\sqrt{x+1} - x + 4 \ln |1-\sqrt{1+x}| + C. \end{aligned}$$

注: 当被积函数中含有 $\sqrt{a^2 - x^2}$, $\sqrt{x^2 \pm a^2}$ 时,
常用三角函数代换法.

例 22. 计算 $\int \sqrt{a^2 - x^2} \, dx \, (a > 0)$.

解: $\int \sqrt{a^2 - x^2} \, dx$

$$\stackrel{x=a \sin t}{=} \int \sqrt{a^2 - a^2 \sin^2 t} \, d(a \sin t)$$

$|t| \leq \frac{\pi}{2}$

$$= \int (a \cos t) \cdot (a \cos t) \, dt = a^2 \int \cos^2 t \, dt$$

$$= a^2 \int \frac{1 + \cos 2t}{2} \, dt = \frac{a^2}{2} \left(t + \frac{\sin 2t}{2} \right) + C$$

$$= \frac{a^2}{2} \left(t + \sin t \cos t \right) + C$$

$$= \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + C.$$

例 23. 计算 $\int \frac{dx}{\sqrt{x^2+a^2}}$ ($a > 0$).

解:
$$\begin{aligned} & \int \frac{dx}{\sqrt{x^2+a^2}} \stackrel{x=a \tan t}{=} \int_{|t| < \frac{\pi}{2}} \frac{d(a \tan t)}{\sqrt{a^2 \tan^2 t + a^2}} \\ &= \int \frac{\cos t dt}{\cos^2 t} = \int \frac{dt}{\cos t} \\ &= \ln |\sec t + \tan t| + C_1 \text{ (见例 18)} \\ &= \ln \left| \frac{1}{a} \sqrt{x^2 + a^2} + \frac{x}{a} \right| + C_1 \\ &= \ln |x + \sqrt{x^2 + a^2}| + C. \end{aligned}$$

例 24. 计算 $\int \frac{1-x+x^2}{\sqrt{1+x-x^2}} dx$.

解: $\int \frac{1-x+x^2}{\sqrt{1+x-x^2}} dx = \int \frac{1-x+x^2}{\sqrt{(\frac{\sqrt{5}}{2})^2 - (x-\frac{1}{2})^2}} dx$

$$\begin{aligned} & \int_{|t| \leq \frac{\pi}{2}} \frac{1 - (\frac{1}{2} + \frac{\sqrt{5}}{2} \sin t) + (\frac{1}{2} + \frac{\sqrt{5}}{2} \sin t)^2}{\sqrt{(\frac{\sqrt{5}}{2})^2 - (\frac{\sqrt{5}}{2} \sin t)^2}} d\left(\frac{1}{2} + \frac{\sqrt{5}}{2} \sin t\right) \\ & x = \frac{1}{2} + \frac{\sqrt{5}}{2} \sin t \end{aligned}$$

$$= \int \left(\frac{3}{4} + \frac{5}{4} \sin^2 t\right) dt = \int \left(\frac{3}{4} + \frac{5}{4} \frac{1 - \cos 2t}{2}\right) dt$$

$$= \frac{11}{8} t - \frac{5}{16} \sin 2t + C = \frac{11}{8} t - \frac{5}{8} \sin t \cos t + C$$

$$= \frac{11}{8} \arcsin \frac{2}{\sqrt{5}} \left(x - \frac{1}{2}\right) - \frac{5}{8} \cdot \frac{2}{\sqrt{5}} \left(x - \frac{1}{2}\right) \cdot \sqrt{1 - \left(\frac{2}{\sqrt{5}} \left(x - \frac{1}{2}\right)\right)^2} + C$$

$$= \frac{11}{8} \arcsin \frac{2}{\sqrt{5}} \left(x - \frac{1}{2}\right) - \frac{1}{2} \left(x - \frac{1}{2}\right) \sqrt{1 + x - x^2} + C.$$

例 25. 计算 $\int \frac{dx}{\sqrt{x^2 - a^2}}$ ($a > 0$).

解: 被积函数的定义域为 $(-\infty, -a) \cup (a, +\infty)$.

当 $x > a$ 时, 考虑变换 $x = a \sec t$ ($0 < t < \frac{\pi}{2}$),

则 $dx = a \frac{\sin t}{\cos^2 t} dt$, 由此可得

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{1}{a \tan t} \cdot a \frac{\sin t}{\cos^2 t} dt = \int \frac{dt}{\cos t} \\ &= \ln |\sec t + \tan t| + C_1 = \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C_1 \\ &= \ln |x + \sqrt{x^2 - a^2}| + C_1'. \end{aligned}$$

当 $x < -a$ 时, 考虑变换 $u = -x$, 则有

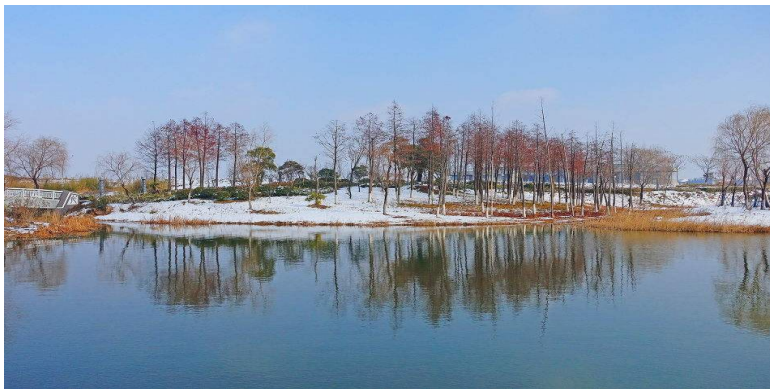
$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - a^2}} &= - \int \frac{du}{\sqrt{u^2 - a^2}} = -\ln |u + \sqrt{u^2 - a^2}| + C_2 \\ &= \ln \left| \frac{1}{-x + \sqrt{x^2 - a^2}} \right| + C_2 = \ln |x + \sqrt{x^2 - a^2}| + C'_2.\end{aligned}$$

于是我们有

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \begin{cases} \ln |x + \sqrt{x^2 - a^2}| + C'_1, & \text{若 } x > a, \\ \ln |x + \sqrt{x^2 - a^2}| + C'_2, & \text{若 } x < -a. \end{cases}$$

因为原函数的定义域由两个不相交的区间组成, 故常数 C_1 和 C_2 可以不同, 但计算不定积分的目的只是为了得到一个原函数, 因此人们通常将上式合并成一个统一的表达式:

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C.$$



同学们辛苦了!